

3. Research Papers (2021 Onward)

The following table lists relevant peer-reviewed and IEEE/Springer/Elsevier/MDPI-indexed papers identified through targeted search. Access some via institutional/library subscription where indicated; open-access links are provided where available.

3.1 Smart Helmets & IoT Safety Systems

Title / Authors (abridged)	Venue, Year	Link / DOI
Advanced IoT-Based Smart Mining Helmet with LoRa-Enabled Communication for Enhanced Safety	IJRASET, 2024	https://www.ijraset.com/research-paper/advanced-iot-based-smart-mining-helmet-with-lora
LoRaWAN Based Smart Safety Helmet with Protection Mask for Miners (Shivaanivarsha et al.)	IEEE IC3IoT, Chennai, 2024	IEEE Xplore (search title)
Smart Helmet and Monitoring for Miners with Enhanced Protection (Banik, Mishra, Manikandan)	IEEE ICECAA, Namakkal, 2023	IEEE Xplore (search title)
Power Efficient Intelligent Helmet for Coal Mining Security and Alerting (Banu et al.)	IEEE ICSCSS, Coimbatore, 2023	IEEE Xplore (search title)
A Protection Approach for Coal Miners Safety Helmet Using IoT (Modi, Mali, Sharma et al.)	Springer, Semantic Intelligence, 2024	https://doi.org/10.1007/978-981-97-3690-4 (chapter 34 series)
IoT-Enabled Safety Helmet for Coal Miners: Real-Time Monitoring, Data Collection, and Analysis	IEEE Conference Publication, 2024	https://ieeexplore.ieee.org/iel8/11026116/11026063/11026450.pdf
Development of an Integrated Safety Monitoring System for Coal Miners using Smart Helmets	IEEE Conference Publication, 2024	https://ieeexplore.ieee.org/iel8/11034706/11034776/11035664.pdf
IoT based Smart Security Helmet for Miner's Safety	IEEE Xplore, 2023	https://ieeexplore.ieee.org/document/10146728/
IoT-Based Smart Helmet for Hazard Detection in Mining Industry	arXiv preprint, 2023	https://arxiv.org/pdf/2304.10156

3.2 PPE Detection Using AI / Computer Vision

Title / Authors (abridged)	Venue, Year	Link / DOI
An effective deep learning approach enabling miners' protective equipment detection and tracking using improved YOLOv7 architecture	ScienceDirect (Computers & Electrical Engineering), 2025	https://www.sciencedirect.com/science/article/abs/pii/S0045790625001168
Real-Time Personal Protective Equipment Compliance Detection Based on Deep Learning Algorithm (Lo, Lin, Hung)	MDPI Sustainability, 2023	https://doi.org/10.3390/su15010391
Personal Protective Equipment Detection: A Deep-Learning-Based Sustainable Approach	MDPI Sustainability, 2023	https://www.mdpi.com/2071-1050/15/18/13990
PPE detector: A YOLO-based architecture to detect personal protective equipment (PPE) for construction sites (Ferdous & Ahsan)	PeerJ Computer Science, 2022	https://doi.org/10.7717/peerj-cs.999 (PeerJ)
Automatic PPE Monitoring System for Construction Workers Using YOLO Algorithm Based on Deep Reinforcement Learning	ResearchGate / conference, 2024	https://www.researchgate.net/publication/386252452
Fast Personal Protective Equipment Detection for Real Construction Sites Using Deep Learning Approaches	PMC / journal article, 2021	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8156681/
Enhancing Construction Worker Safety through YOLOv8-Based PPE Detection	IJRASET, 2024	https://www.ijraset.com/research-paper/enhancing-construction-worker-safety-through-yolov8-based-ppe-detection

3.3 IoT Health Monitoring & Wearable Sensors

Title / Authors (abridged)	Venue, Year	Link / DOI
Healthcare Monitoring Using an Internet of Things-Based Cardio System	MDPI IoT journal, 2025	https://www.mdpi.com/2624-831x/6/1/10
Evaluating the Accuracy of Low-Cost Wearable Sensors for Healthcare Monitoring (nRF52840, MAX32664D-based)	MDPI Sensors, 2025	https://www.mdpi.com/2072-666X/16/7/791
A Wearable Internet of Things Device for Noninvasive Remote Monitoring of Vital Signs Related to Heart Failure	MDPI IoT journal, 2024	https://doi.org/10.3390/iot5010008
Recent Advances on IoT-Assisted Wearable Sensor Systems for Healthcare Monitoring	PMC / MDPI Sensors review	https://pmc.ncbi.nlm.nih.gov/articles/PMC8534204/
A Comprehensive Survey on Wearable Computing for Mental and Physical Health Monitoring	MDPI Electronics, 2025	https://www.mdpi.com/2079-9292/14/17/3443

3.4 Underground Localization & Navigation

Title / Authors (abridged)	Venue, Year	Link / DOI
Research on IMU-Assisted UWB-Based Positioning Algorithm in Underground Coal Mines	MDPI Micromachines / PMC, 2023	https://pmc.ncbi.nlm.nih.gov/articles/PMC10384321/
Accurate integrated position measurement system for mobile applications in GPS-denied coal mine	ScienceDirect (Measurement), 2023	https://www.sciencedirect.com/science/article/abs/pii/S0019057823001830
Development and Evaluation of a UWB-Based Indoor Positioning System for Underground Mine Environments	Mining, Metallurgy & Exploration (Springer), 2024	https://doi.org/10.1007/s42461-023-00797-z
UWB-Based Localization System Aided With Inertial Sensor for Underground Coal Mine Applications	IEEE Journals & Magazine	https://ieeexplore.ieee.org/abstract/document/9007738
Underground mine positioning: A review (Seguel, Palacios-Játiva, Azurdia-Meza et al.)	IEEE Sensors Journal, 2022, 22(6), 4755–4771	IEEE Xplore (search title)
Human Real Time Localization System in Underground Mines using UWB	ResearchGate / conference paper	https://www.researchgate.net/publication/341036813

Note: several IEEE Xplore entries above are best accessed directly via IEEE Xplore search using the exact title, since IEEE assigns DOIs at the article/session level that can shift between preprint and final publication; searching the title on ieeexplore.ieee.org will resolve to the authoritative page and citation-ready DOI.